

Requirement Analysis

Phase 2 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Functional Requirements

- The system shall accept Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall as numeric input through a web form.
- The system shall pass the validated input to a trained Logistic Regression model and return the single best-suited crop.
- The system shall display the recommended crop clearly on the Find Your Crop page.
- The system shall provide an About page describing the project's purpose and methodology.
- The system shall support navigation between Home, About, and Find Your Crop pages via a persistent top navigation bar.

Non-Functional Requirements

- Usability: input fields should be clearly labeled with the expected agricultural parameter, so farmers or agricultural extension workers can fill the form without technical training.
- Performance: crop recommendations should return in under one second given the lightweight Logistic Regression model.
- Reliability: the system should handle unexpected or missing input gracefully without crashing the Flask server.
- Scalability: the underlying pipeline should support retraining on larger or updated agricultural datasets without changes to the web application code.

Business Requirements

The platform must process key soil and climate parameters (N, P, K, temperature, humidity, rainfall, pH) to identify the most suitable crop for cultivation, helping optimize the usage of water, fertilizers, and soil nutrients while reducing the risk of crop failure caused by poor crop selection.

Hardware and Software Requirements

Category	Requirement
Operating System	Windows 10/11, macOS, or Linux
Python Version	Python 3.9 or higher

Category	Requirement
Core Libraries	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
Web Framework	Flask
Development Tools	Anaconda Navigator / PyCharm / VS Code, Jupyter Notebook
Minimum RAM	4 GB (8 GB recommended)
Browser	Any modern browser to access the web app